EE5203 L01 Embedded C Readiness Diagnostic

Time: 15 minutes
Resources: None
Grading: Diagnostic only; results select the appropriate C practice track.

Part A - Trace familiar programming ideas

1. What is the final value of `x`?

```c
int x = 3;
x = x + 2;
x = x * 2;
```

2. What is the final value of `status`?

```c
int temperature = 18;
int status = 0;

if (temperature > 20) {
    status = 1;
}
```

3. List the values printed or observed during this loop.

```c
for (int i = 0; i < 4; i++) {
    /* observe i */
}
```

4. Complete the function so it returns the larger input.

```c
int maximum(int a, int b)
{
    /* your code */
}
```

5. The array contains four samples:

```c
int samples[4] = {10, 20, 30, 40};
```

a. What is `samples[0]`?

b. What is `samples[3]`?

c. Which index would be invalid?

Part B - Starting representation knowledge

6. Write decimal 13 in binary.
7. Write binary `1010` in decimal.
8. Which hexadecimal digit represents decimal 15?
9. Without calculating the final numeric value, describe what this expression does to bit 3 of `value`:

```c
value | (1U << 3)
```

Choose one: **sets it / clears it / toggles it / not sure**.

Part C - Debugging

10. A compiler reports errors on lines 8, 12, and 20. Which error should normally be investigated first, and why?
11. In one sentence, distinguish a syntax error from a logic error.